

Skill-1

1. To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile

```
gowtham@DESKTOP-2F03K3T:~$ sudo apt update
[sudo] password for gowtham:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:5 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [383 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [458 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [92.0 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [112 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.6 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.8 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [154 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [237 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [77.8 kB]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [50.0 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [182 kB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [341 kB]
Get:18 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [324 kB]
Get:19 http://security.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [65.7 kB]
Get:19 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [65.7 kB]
Get:20 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [61.6 kB]
Get:21 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [11.3 kB]
Get:22 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [3032 B]
```

Fetched 3016 kB in 7s (407 kB/s)

20 packages can be upgraded. Run 'apt list --upgradable' to see them.

```
gowtham@DESKTOP-2F03K3T:~$
```

```
gowtham@DESKTOP-2F03K3T:~$ sudo apt update
Upgrading:
  alsa-ucm-conf      libjavascriptcoregtk-6.0-1  libudev1      systemd-cryptsetup
  apparmor           libnss-systemd              libwpebackend-fdo-1.0-1  systemd-resolved
  cloud-init         libpam-systemd              libyelp-1-0      systemd-sysv
  cloud-init-base    libsystemd-shared           linux-libc-dev    udev
  libapparmor1       libsystemd0                 systemd          yelp

Summary:
  Upgrading: 20, Installing: 0, Removing: 0, Not Upgrading: 0
  13 standard LTS security updates
  Download size: 45.2 MB
  Space needed: 42.0 kB / 1023 GB available

Continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libsystemd0 amd64 259.5-0ubuntu3.4 [481 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 systemd-sysv amd64 259.5-0ubuntu3.4 [9250 B]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 systemd-cryptsetup amd64 259.5-0ubuntu3.4 [1299 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libpam-systemd amd64 259.5-0ubuntu3.4 [274 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libsystemd-shared amd64 259.5-0ubuntu3.4 [2462 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 systemd amd64 259.5-0ubuntu3.4 [3129 kB]
11% [6 systemd 15.6 kB/3129 kB 0%] 576 kB/s 1min 12s
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 systemd-cryptsetup amd64 259.5-0ubuntu3.4 [1244 kB]
18% [7 systemd-cryptsetup 14.2 kB/124 kB 11%] 576 kB/s 1min 6s
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 udev amd64 259.5-0ubuntu3.4 [1501 kB]
20% [8 udev 308 kB/1501 kB 21%] 576 kB/s 1min 5s
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libudev1 amd64 259.5-0ubuntu3.4 [151 kB]
24% [Working] 576 kB/s 1min 3s
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libapparmor1 amd64 5.0.2-0ubuntu1-26.04.1 [51.3 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libnss-systemd amd64 259.5-0ubuntu3.4 [227 kB]
26% [Working] 576 kB/s 1min 3s
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 apparmor amd64 5.0.2-0ubuntu1-26.04.1 [380 kB]
27% [12 apparmor 1156 B/380 kB 0%] 576 kB/s 1min 3s
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 alsa-ucm-conf all 1.2.15.3-1ubuntu1.5 [106 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 cloud-init-base all 26.1-0ubuntu3-26.04.1 [587 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 cloud-init all 26.1-0ubuntu3-26.04.1 [2110 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libjavascriptcoregtk-6.0-1 amd64 2.46.5-0ubuntu0.26.04.1 [7282 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libwpebackend-fdo-1.0-1 amd64 1.15.2-0ubuntu1-26.04.1 [323 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libyelp-1-0 amd64 42.2-0ubuntu1-26.04.1 [119 kB]
Get:19 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 yelp amd64 42.2-0ubuntu1-26.04.1 [734 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-libc-dev amd64 6.14.0-15.15-26.04.1 [1553 kB]
Get:22 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libsystemd0 amd64 259.5-0ubuntu3.4 [2559 kB]
Fetched 45.2 MB in 7s (6606 kB/s)
20 packages can be upgraded. Run 'apt list --upgradable' to see them.
gowtham@DESKTOP-2F03K3T:~$
```

```
gowtham@DESKTOP-2F03K3T:~/OS$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
gowtham@DESKTOP-2F03K3T:~/OS$
```

```
gowtham@DESKTOP-2F03K3T:~/OS$ nano hello.c
gowtham@DESKTOP-2F03K3T:~/OS$ gcc hello.c -o hello
gowtham@DESKTOP-2F03K3T:~/OS$ ./hello

Parent Process Started
Parent PID : 3152
Parent PPID : 401

Parent created Child
Child PID : 3153
Child Process Started
Child PID : 3153
Child PPID : 3152

Child is executing 'ls -l' using exec()

total 264
drwxr-xr-x 6 gowtham gowtham 4096 Aug 1 08:35 GIT
drwxr-xr-x 2 gowtham gowtham 4096 Aug 5 07:10 OSSP
drwxr-xr-x 2 gowtham gowtham 4096 Aug 3 05:55 Skill1
drwxr-xr-x 2 gowtham gowtham 4096 Aug 3 05:32 Skill10
drwxr-xr-x 2 gowtham gowtham 4096 Aug 3 05:32 Skill11
drwxr-xr-x 2 gowtham gowtham 4096 Aug 3 05:32 Skill12
drwxr-xr-x 2 gowtham gowtham 4096 Aug 3 05:32 Skill13
drwxr-xr-x 2 gowtham gowtham 4096 Aug 3 05:32 Skill14
drwxr-xr-x 2 gowtham gowtham 4096 Aug 3 05:32 Skill15
drwxr-xr-x 2 gowtham gowtham 4096 Aug 3 05:32 Skill16
drwxr-xr-x 2 gowtham gowtham 4096 Aug 3 05:32 Skill17
drwxr-xr-x 2 gowtham gowtham 4096 Aug 3 05:32 Skill18
drwxr-xr-x 2 gowtham gowtham 4096 Aug 3 05:32 Skill19
-rw-r--r-- 1 gowtham gowtham 16552 Aug 8 09:21 client
-rw-r--r-- 1 gowtham gowtham 1348 Aug 8 09:21 client.c
-rw-r--r-- 1 gowtham gowtham 16056 Jul 28 03:43 file
-rw-r--r-- 1 gowtham gowtham 226 Jul 28 03:43 file.c
drwxr-xr-x 2 gowtham gowtham 4096 Jul 31 15:48 fork
-rw-r--r-- 1 gowtham gowtham 16288 Aug 14 08:59 hello
-rw-r--r-- 1 gowtham gowtham 986 Aug 14 08:59 hello.c
-rw-r--r-- 1 gowtham gowtham 16264 Aug 8 08:37 practical2
-rw-r--r-- 1 gowtham gowtham 1420 Aug 8 08:52 practical2.c
-rw-r--r-- 1 gowtham gowtham 1218 Aug 8 08:48 practical3
-rw-r--r-- 1 gowtham gowtham 16344 Aug 8 08:56 practical4
-rw-r--r-- 1 gowtham gowtham 1787 Aug 8 08:59 practical4.c
-rw-r--r-- 1 gowtham gowtham 16432 Aug 8 09:15 practical5
-rw-r--r-- 1 gowtham gowtham 1548 Aug 8 09:13 practical5.c
-rw-r--r-- 1 gowtham gowtham 16136 Aug 8 08:54 process_states
-rw-r--r-- 1 gowtham gowtham 413 Aug 8 08:54 process_states.c
-rw-r--r-- 1 gowtham gowtham 24 Aug 8 08:40 sample.txt
-rw-r--r-- 1 gowtham gowtham 16536 Aug 8 09:21 server
-rw-r--r-- 1 gowtham gowtham 1728 Aug 8 09:20 server.c
-rw-r--r-- 1 gowtham gowtham 40 Aug 8 08:38 source.txt
-rw-r--r-- 1 gowtham gowtham 16208 Aug 8 09:05 zombie
-rw-r--r-- 1 gowtham gowtham 698 Aug 8 09:05 zombie.c

Child process completed.
Parent process completed.
gowtham@DESKTOP-2F03K3T:~/OS$
```

```
gowtham@DESKTOP-2F03K3T:~/OS$ mkdir kernel-lab
gowtham@DESKTOP-2F03K3T:~/OS$ cd kernel-lab
gowtham@DESKTOP-2F03K3T:~/OS/kernel-lab$ mkdir src include bin
gowtham@DESKTOP-2F03K3T:~/OS/kernel-lab$ touch README.md
Makefile
gowtham@DESKTOP-2F03K3T:~/OS/kernel-lab$ ls -R
.:
Makefile  README.md  bin  include  src

./bin:

./include:

./src:
gowtham@DESKTOP-2F03K3T:~/OS/kernel-lab$
```